- Meiosis II separates the sister chromatids.
- Sister chromatid cohesion and crossing over allow chiasmata to hold homologs together until anaphase I. Cohesins are cleaved along the arms at anaphase I, allowing homologs to separate, and at the centromeres in anaphase II, releasing sister chromatids from each other.

? *In prophase I, homologous chromosomes pair up and undergo synapsis and crossing over. Can this also occur during prophase II? Explain.*

CONCEPT **13.4**

Genetic variation produced in sexual life cycles contributes to evolution *(pp. 265–267)*

- Three events in sexual reproduction contribute to genetic variation in a population: independent assortment of chromosomes during meiosis I, crossing over during meiosis I, and random fertilization of egg cells by sperm. During crossing over, DNA of nonsister chromatids in a homologous pair is broken and rejoined.
- Genetic variation is the raw material for evolution by natural selection. Mutations are the original source of this variation; recombination of variant genes generates additional genetic diversity.

? *Explain how three processes unique to sexual reproduction generate a great deal of genetic variation.*

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. A human cell containing 22 autosomes and a Y chromosome is
(A) a sperm. (C) a zygote.
(B) an egg. (D) a somatic cell of a male.

2. The two homologs of a pair move toward opposite poles of dividing cell during
(A) mitosis. (C) meiosis II.
(B) meiosis I. (D) fertilization.

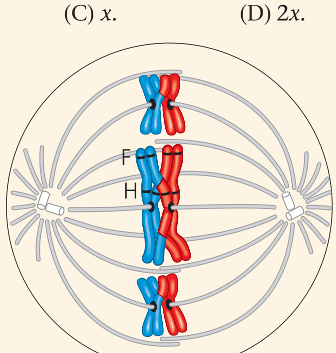
Levels 3-4: Applying/Analyzing

3. Meiosis II is similar to mitosis in that
(A) sister chromatids separate during anaphase.
(B) DNA replicates before the division.
(C) the daughter cells are diploid.
(D) homologous chromosomes synapse.

4. If the DNA content of a diploid cell in the G_1 phase of the cell cycle is x, then the DNA content of the same cell at metaphase of meiosis I will be
(A) $0.25x$. (B) $0.5x$. (C) x. (D) $2x$.

5. If we continue to follow the cell lineage from question 4, then the DNA content of a single cell at metaphase of meiosis II will be
(A) $0.25x$. (B) $0.5x$. (C) x. (D) $2x$.

6. DRAW IT The diagram shows a cell in meiosis.



(a) Label the appropriate structures with these terms: chromosome (label as duplicated or unduplicated), centromere, kinetochore, sister chromatids, nonsister chromatids, homologous pair (use a bracket when labeling), homolog (label each one), chiasma, sister chromatid cohesion, and gene loci. Circle and label the alleles of the F gene.
(b) Precisely describe the makeup of a haploid set and a diploid set in this cell.
(c) Identify the stage of meiosis shown.

Levels 5-6: Evaluating/Creating

7. Explain how you can tell that the cell in question 6 is undergoing meiosis, not mitosis.

8. EVOLUTION CONNECTION Many species can reproduce either asexually or sexually. Explain what you think might be the evolutionary significance of the switch from asexual to sexual reproduction that occurs in some species when the environment becomes unfavorable.

9. SCIENTIFIC INQUIRY The diagram in question 6 represents just a few of the chromosomes of a meiotic cell in a certain person. Assume that the freckles gene is located at the locus marked F, and the hair-color gene is located at the locus marked H, both on the long chromosome. The individual from whom this cell was taken has inherited different alleles for each gene ("freckles" and "black hair" from one parent, and "no freckles" and "blond hair" from the other). Predict allele combinations in the gametes resulting from this meiotic event. (It will help if you draw out the rest of meiosis and label the alleles by name.) List other possible combinations of these alleles in this individual's gametes.

10. WRITE ABOUT A THEME: INFORMATION The continuity of life is based on heritable information in the form of DNA. In a short essay (100–150 words), explain how chromosome behavior during sexual reproduction in animals ensures perpetuation of parental traits in offspring and, at the same time, genetic variation among offspring.

11. SYNTHESIZE YOUR KNOWLEDGE

The Cavendish banana, the world's most popular fruit, is currently threatened by extinction due to a fungus. This banana variety is "triploid" ($3n$, with three sets of chromosomes) and can only reproduce through cloning by cultivators. Given what you know about meiosis, explain how the banana's triploid number accounts for its inability to form normal gametes. Considering genetic diversity, discuss how the absence of sexual reproduction might make this domesticated species vulnerable to infectious agents.

For selected answers, see Appendix A.